

Economic Instability, Market Opening and Adjustment Strategy in the Brazilian Industry

Francisco Lima C. Teixeira^{*}
Joao Carlos Ferraz^{**}

1. Introduction

Following the external debt crisis of 1982, the Brazilian economy entered a period of profound macroeconomic instability, in spite of successive economic plans and policies aimed at stabilization. Indeed, from 1980 to 1994, Brazil accumulated nine stabilization plans, 15 different wage policies, 19 changes in the exchange regime, 22 proposals for dealing with the external debt and 20 attempts for adjusting public finances. Economic stagnation and industrial decline also characterize this period, after 50 years of outstanding industrial development. From the 1982 debt crisis onwards, Brazilian industry lagged behind the international trajectory towards technological and managerial changes. The old *Import Substitution* strategy could no longer provide the grounds for economic and industrial growth.

From the end of the 80's, a new development strategy, named *Competitive Integration*, started to emerge. The main idea underlying this was that, in order to enter a new period of development, Brazil had to open its economy and try to participate in the *globalization* process. In practice, this meant trade liberalization, deregulation and decreasing state intervention. The year of 1990 represents a landmark in the adoption of the *Competitive Integration* policy. Since then, trade liberalization has been increased, privatization carried out and free market mechanisms enforced, exposing the local industry to full

^{*} From Federal University of Bahia, Brasil. Presently in a Visiting Program in the University of Texas, Austin.

international competition. However, stability and economic growth were only partially achieved with the *Plano Real* of 1994.

Throughout this process, Brazilian industry has developed a varied adjustment behavior in order to cope with internal macroeconomic changes and external competition. The objective of this paper is to discuss the impacts, on the structure of Brazilian industry, of the adjustment process it has been through since the early eighties. In order to do that, first, a brief characterization of the institutional changes and the macroeconomic background against which the adjustment process took place is presented. Second, the adjustment strategies adopted by firms aiming at surviving under a hostile environment are reviewed and their main results and problems analyzed. Finally, based on exports and imports time series and on investment behavior data, the impacts of these changes on the structure of Brazilian industry and its future prospects are to be discussed.

2. Institutional changes and macroeconomic background

The main features of the institutional changes, which implied the replacement of the *Import Substitution* for the *Competitive Integration* strategy, are well known (see, for instance, Miranda, 1994). In a few words, from a macroeconomic perspective, these changes involved adoption of the so-called *Washington consensus* (Williamson, 1993), or, as Krugman (1996) puts it, *the new conventional wisdom*: Free trade and attempts to impose tight fiscal, monetary and exchange policies. However, alternatively, these changes can be seen as composed of three main blocks: **market focus**, **industrial structure** and **regime of incentives and regulations**.

During the *Import Substitution* period, the **market** was viewed essentially as formed by the local demand, which should be supplied from inside the country.

** From Federal University of Rio de Janeiro, Brazil, and University of Tsukuba, Japan.

The **industrial structure** was mainly based on state and foreign companies, with local private companies playing a minor role, and the main goal was to establish production capabilities. The regime of **incentives and regulations** was characterized by market protection, incentives for nationalization of production and strong government intervention. The final goal of the industrialization process, in the Brazilian case, seemed to be the achievement of a sustainable self-sufficiency in production, by means of fast industrial growth.

The *Competitive Integration* strategy can also be named the *market friendly approach* to industrialization. Trade liberalization, aiming at increasing both imports and exports and exposing the local industry to international competition, has changed the **market focus**. The **industrial structure**, in turn, was set to be modified by means of privatization of state companies and attraction of foreign direct investment. Decreasing government intervention in the economy (deregulation) has been the main feature of the regime of **incentives and regulations**. The priority to growth was replaced by the concern with macroeconomic fundamentals which, until the *Plano Real*, did not ensure fiscal, monetary and exchange stability

Therefore, these institutional changes were carried out under a very unstable macroeconomic background. In fact, in spite of the profusion of attempts, until the 1994 *Plano Real*, Brazilian economy had experienced a permanent inflationary process with strong trends towards open hyperinflation, only contained, momentarily, by increasingly ineffective stabilization plans. Inflation rates, measured by different price indexes, presented very abrupt oscillations and, from 1988 onwards, four digit figures were quite common. In 1989, the general price index went above 1,800% and, in 1993, above 2,400%. No other country experienced such a long period of high and chronic inflation and went further in promoting the indexation of its economy than Brazil.

The inflationary process, together with the costs of adjusting to the external debt, resulted in a deep crisis of the public sector, previously the main source of new investment in the economy. Also, due to the lack of feasible interest rates, which could compensate for the risks of the economic agents, Brazilian economy gradually drifted to a state of financial regression. The disarray of the financial system led to a drastic and permanent fall in the demand for consumer and investment credit.

The results of this instability process can be seen in the economic aggregates presented in Table 1. In spite of some years of erratic growth (like 1985), GDP per capita in 1994 was still below that of 1980. As demand for goods fell sharply, firms invested as little as possible, limiting equipment renewal, which implied growing industrial obsolescence. The investment rate fell from 22.8% in 1980 to 14.5% in 1992, and has not been recovered by 1995.

An overall balance of this long period of instability indicates the destruction of a development paradigm - Import Substitution - in which Brazil, among all other developing countries, seemed to be the most successful case. However, the transition to a new development paradigm has proved to be a long and costly process. Public and private institutions have not been able to change objectives and implement a regime of incentives and regulations suitable for ensuring sustainable development, especially industrial growth

Table 1: Brasil: Evolution of economic aggregates
(1980-1995)

Year	Growth Rate (%)	GDP per capita (US\$)	Investment/GDP (%)
1980	9.2	3,552	22.8
1981	-4,5	2,958	20,9
1982	0,3	2,916	19,4
1983	-3,3	2,761	16,9
1984	5,2	2,852	16,2
1985	7.9	3,399	16.3
1986	7,6	3,187	18,7
1987	3,5	3,239	17,8
1988	0,0	3,174	17,0
1989	3,2	3,217	16,5
1990	-4.4	3,402	15.8
1991	1.2	3,359	15.1
1992	-0.7	3,280	14.5
1993	4.8	3,368	15.0
1994	5.8	3,512	15.0
1995	4.2	3,609	15.4

Source: GDP and GDP per capita: Banco Central do Brazil, Annual reports, values at 1995 prices. Investment - revised figures by IPEA in accordance with a new methodology defined by IBGE, in: Indicadores IESP, nº 28, maio 1994, pg. 44.

3. Adjustment strategies

The adjustment strategies followed by the Brazilian industry can be viewed as having two main phases, each one with two periods, as presented in Figure 1. Few reminders are necessary for a correct understanding of this Figure. First, this is not a periodization chart: Dates indicate the approximate start period for each type of behavior. Initiated by leader firms, these behaviors are later diffused within industry at increasing rates, leveling off if and when most firms adopt them. Second, and more important, the net consequences of each behavior are cumulatively carried on to the following years, e.g. once adopted, the preference for liquidity is not easily abandoned by firms. Finally, each type of behavior has a particular time span and involves a specific group of firms; e.g. not all Brazilian firms turned to the export markets. That is to say, at any given moment of time,

firms with different attributes -size, sector of origin, etc.- could be evolving at varying rates and following a path divergent from the most common orientation. The adoption of these strategies implied an *economic premia* (Fanelli and Frenkel, 1994), in the sense that it has given capable firms the power to facing uncertainty and expanding production when stabilization and growth have eventually been reached.

Figure 1: Adjustment Strategies by Brazilian Firms

Phases	Periods	Macroeconomic determinants	Type of strategy	General orientation	Microeconomic implications
I	1981/82	Debt crisis and the need for hard currency	Export adjustment	Expand exports and local production	Market broadening
	1982/88	Economic instability with spells of inflation control	Financial adjustment	High liquidity, low debt-equity ratios	Reduced financial exposure
II	1988/93	Growing economic instability, recession and trade liberalization	Cost adjustment	Cost rationalization, downsizing, imports of components	Productivity increases
	1994-	Monetary stabilization with stop and go growth in demand	Expansion adjustment	Mergers, optimization of installed capacity, product specialization	Fast Restructuring

Source: Adapted from Ferraz, Kupfer and Serrano (1997)

The first phase corresponds to the export adjustment of the Brazilian economy, in response to the debt crisis, rooted in the investment drive of the late seventies (Castro and Pires de Souza, 1985). In the first period of this phase (1980-82), the main government policy that affected industry was the granting of fiscal and credit subsidies for exporters, combined with real exchange devaluation. At the same time, imports were compressed even further than the previous low levels.

A large set of firms with production and marketing capacity tried to adjust to this situation by exploring the external market, in a type of **export adjustment strategy**. In doing so, they acquired broad market view and flexibility

In the second period of phase I, amid plans aiming at curbing inflation, the Brazilian government assumed responsibility over external debt incurred by private and state companies, as a means of providing a buffer for an overindebted private sector against currency devaluation and increases in international interest rates. Firms could pay their debt in local currency, rather than paying back international banks in foreign exchange.

This policy measure, coupled with persistent instability, led firms to pursue high liquidity and to cut their debt exposure behavior, characterizing a **financial adjustment strategy**. Permanent price instability and institutional distortions created by the various and ill-fated attempts to curb down inflation shortened decision horizons. Productive investment plans were shelved and replaced by an inflation-hedging behavior. This preference for liquidity became anchored in very sophisticated indexation systems, beating government attempts to extinguish them. Consequently, whereas, in 1982, the 1,000 largest industrial firms had a debt equity ratio of 54.6 and a liquidity of 1.2, by 1988, the debt equity ratio was 34.1 and liquidity 1.5 (Ferraz, Rush and Miles, 1992). At this time, financial profits were, generally, higher than operational ones.

The second phase of the adjustment process corresponds to the introduction of trade liberalization. Since 1988, a tariff reform was being implemented, involving removal of non-tariff barriers and a reduction in the average manufacturing tariff from 90 to 43%. In 1990, trade liberalization was accelerated, including the removal of most non-trade barriers and a four year advance schedule for tariff reductions to 20% by 1994. In the early 1990s, international lending constraints improved and financial inflows increased drastically, attracted by rates of returns similar to other newly *emerging markets*, helping to finance the import bill.

However, in spite of new plans by Collor's government, inflation was not reduced to acceptable levels in a sustained basis. These macroeconomic policies provoked an internal recession, reducing even further the internal demand.

For the industrial firms, the first period of this second phase is characterized by the adoption of a **cost adjustment strategy**. After decades of operating under the 'nationalization at any cost' policy, a growing number of firms started efforts to 'produce at minimum cost', a move inaugurated by industrial leaders some few years before. However, what became known as the *Brazilian modernization* was a defensive, cost cutting drive, impelled by economic instability coupled with trade liberalization. The main objective was to maintain established market positions in face of internal recession and external competition.

Work force, administration procedures, production times, waste indexes and input procurement became the areas subjected to rationalization. *Downsizing* became the favorite motto of managers, implying two general movements. First, specialization or reduction in the number of products locally produced, consequently substituting them for imports. Second, deverticalization of production, which also implied the substitution of locally produced components for imports. At the same time, firms tried to upgrade the technological levels of products and to set prices at import level.

Costs adjustments were facilitated by the use of a wide range of generic innovations widely available at low cost, especially microelectronic-based industrial automation and new organizational techniques. Industrial efforts to improve productive efficiency became widely known in industrial circles and in the general media, particularly after 1992, when there was an explosion in the number of firms certified under the ISO 9,000 standards. Generally, ISO certification was associated with Total Quality Management programs.

The implementation of *Plano Real*, in 1994, inaugurated the second period of phase two. This economic plan greatly benefited from the final resolution of the external debt problem, associated with an increasing inflow in foreign exchange into the economy. Since 1992, international interest rates had been falling and investors were being attracted to 'emerging', recently deregulated, markets. As a result, cash international reserves of the Central Bank had increased from US\$ 8.5 billions, in 1991, to US\$ 19 billions, in 1992, and US\$ 36 billions, in 1994, reaching US\$ 55 billions at the end of 1996. (ECLAC 1996). From a monthly level of 47.7%, in June 1994, inflation dropped to 17.9%, annually, in the second half of 1994 and to 10.6% and 7.6%, in the first and second semesters of 1995.

Immediately after stabilization, economic growth resumed fueled, to a great extent, by an outstanding 'income effect'. Price increases were constrained by an overvalued exchange rate and by a surge in imports, supported by higher reserves, as foreign capital was attracted by high interest rates. As a consequence, Brazilian economy showed a 4.2% annual rate of growth in 1993, 5.7% in 1994 and 4.2% in 1995 and 2.9%, in 1996. According to the government research institute IPEA, there was a sharp decline in social inequality, as the 50% poorest gained 1.2 and the 20% richest lost 2.3 percentage points of their participation in total income, after two years of *Plano Real* (MFAZENDA 1996).

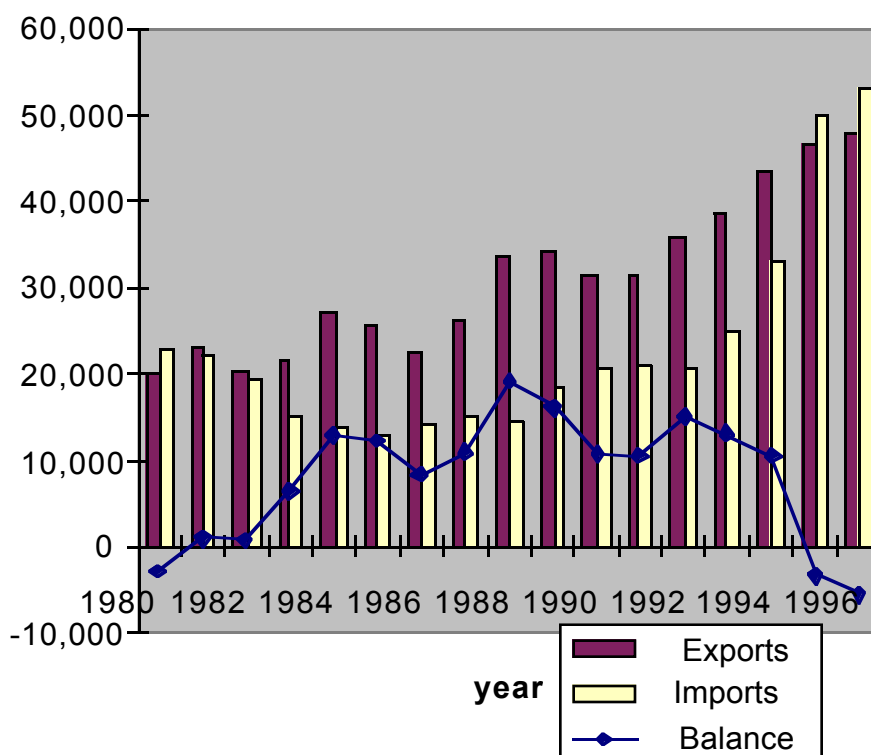
For Brazilian industry the introduction of *Plano Real* represented a change towards an **expansionary adjustment strategy**. This strategy implies product specialization, optimization of installed capacity and restructuring through mergers and acquisitions. The objective was to obtain productive capacity flexibility in order to adjust to internal demand fluctuations and import increases, elevating the competitiveness of firms.

3.1 Results and problems of adjustment

The results and problems resulting from the adjustment process of Brazilian industry are going to be analyzed, first, from a macroeconomic perspective, trying to highlight their impacts on the trade balance. After that, data about changes at the industry level are reviewed.

According to Bielshowsky and Stumpo (1996), as far as trade balance is concerned, since 1980 Brazilian industry's international integration has gone through two phases. The first, during the 1980s, was marked by increases in the export coefficient of the economy. Conversely, during these years, imports were strongly repressed. The second phase, in the 1990s, was marked by a surge in import coefficients. These two phases can be clearly seen in Graph 1.

Graph 1: Brazilian Trade Balance



Source: Banco Central of Brazil

Indeed, up to 1988, trade surpluses were in a crescent trajectory due both to export increases and import decreases. During this period, the external adjustment of the Brazilian economy paid its results, generating enough foreign exchange to cover deficits in services, including interest payments. Subsidies, exchanges rates and internal recession helped the export boom during these years. However, it is clear that the government policies echoed among industrial companies, which were capable to adjust to the new situation. On its turn, the declining import bill has to be appreciated against the lack of external credit, internal recession and the gradual decline in oil prices.

From 1990, the situation of the Brazilian trade balance has started to change. Impacted by trade liberalization, import coefficients increase while exports were not growing at the same rate. A trade deficit period was inaugurated in 1995, reaching U\$ 5,5 billions in 1996 and US\$ 3 billion in the first quarter of 1997. For that, currency valuation, considered to be essential for *Plano Real* success, was of much help. These deficits have been financed by inflows of capital, mainly short-term investment. Nevertheless, the recent trajectory of Brazil's trade deficit is the main economic debate in the country now: Is that possible to finance this increasing bill without threatening the stabilization plan by means of currency devaluation? Or, is Brazilian industry entering a phase of competitiveness improvement that, eventually, will reverse the trade deficits? The analysis that follows tries to outline a possible answer to this second question.

Table 2 presents data about Brazilian industrial evolution from 1986 to 1994. It shows that firms were able to extract higher production levels without expanding employment or the work journey, significantly elevating productivity. However, physical production has not increased at the same rate as productivity. Firms occupied idleness of an extended, optimized production capacity and reaped most benefits as employment and wages did not grow as fast. On the contrary, during recent years, changes in average real wage remained below changes in productivity levels, indicating that the benefits of higher efficiency were being

channeled to the firms. Industrial employment has also shown a declining trend, affecting mainly highly industrialized areas, as Sao Paulo. As potential production levels were above the pre-rationalization years, most pressures for capital investment were cooled down. Such unique feature was extremely functional for the low confidence context still prevailing among the business community in the early 1990s.

The problems associated with the adjustment strategy so far adopted have been mainly related with investment questions. Growth in productivity using installed production capacity has been able to adjust part of Brazilian industry to trade liberalization and international competitiveness. With stabilization, as from 1994, even some GDP growth has been achieved (see Table 1). However, the low level of investment in new productivity capacity raises doubts about the prospects of future economic growth. As can be seen in Table 1, investment as a proportion of GDP in 1995 (15.4%) did not reach the level it had ten years earlier, imposing a physical constraint for further growth: In 1996 GDP growth was only 2,9%.

Table 2: Brazil: Industrial evolution 1986-1994
(previous year = 100)

Year	Physical production	Employment	Work journey	Productivity	Average real wage
1986	110.9	111.0	101.9	98.1	112.5
1987	100.9	101.2	100.7	99.3	93.6
1988	96.8	96.1	100.2	100.6	100.9
1989	102.9	101.8	95.1	105.8	94.1
1990	91.1	94.6	98.5	98.0	87.2
1991	97.4	89.8	100.2	108.0	99.7
1992	96.3	91.2	99.9	106.1	111.6
1993	107.5	97.4	100.1	110.1	106.8
1994	107.5	98.0	99.5	110.4	105.7

Source: IBGE (Brazilian Institute of Geography and Statistics)

Another source of concern exists in relation to technological development. There are no strong signs of an effective inclination, among Brazilian industrial firms,

towards investing in the development of new products and processes. Most available evidence indicates firms moving towards specialization in standardized products with consequent stagnation or decline in R&D expenditure, especially in technology intensive industries, as shown in Table 3. In average, Brazilian firms spend 0.7% of sales in R&D and this level has not changed in the past 5 to 7 years. Besides, while in international technology-based industries 5 to 10% R&D to sales expenditures are common, in Brazil there was a decline from 2.3% to 1.8% between 1987-89 and 1992

Technology investments respond to positive expectations towards the long-term prospects and, in the Brazilian case, this would indicate the beginning of a real transformation on the sources of competitive advantage. The past heritage of low priority to technology investment, aggravated during the adjustment process, has legated an inertia heritage, which may hinder future evolution towards an industrial development centered on innovation, a fundamental condition for international competitiveness.

Table 3: Brazil: R&D and Engineering expenditures
(% of total sales)

Categories	Capital Goods		Commodities		Durable goods		Traditional goods	
	87-89	1992	87-89	1992	87-89	1992	87-89	1992
R & D	2,30	1,77	0,40	0,46	0,66	0,70	0,51	0,59
Engineering	2,05	1,86	1,26	1,94	1,79	2,21	0,52	0,47

Source: Ferraz, Kupfer and Haguenaer (1996)

4. Impacts on the structure of industry

The analysis of the changes in the structure and competitiveness of Brazilian industry is based on foreign trade data presented in Tables 4 and 5. Table 4 contains the Brazilian export figures in three moments: 1965 - when *Import Substitution* was the main policy and the fast growth years were about to start; 1980 - the last year of growth before the debt crisis; and 1995 after the adjustment process has gone deeply in Brazilian industry.

The main general trends are, first, a remarkable decrease of primary goods participation in total exports (from 68% in 1965 to 14% in 1995), with a correspondent increase in industrial goods participation (from 23 to 74% in the same period). These changes reflect the industrialization process under *Import Substitution* strategy. Second, export of services increased only marginally, from 9% of total exports, in 1965, to 12%, in 1995.

Table 4: Brazil: Exports 1965/1995
(1995 US\$ millions and percentages)

Exports	1965		1980		1995	
	Value	%	Value	%	Value	%
Primary goods	1,188	68	5,824	27	7,386	14
Industrial goods	407	23	14,308	65	38,760	74
traditional*	308	76	7,280	51	14,533	38
commodities*	68	17	3,500	25	14,765	38
durable*	8	2	1,358	9	3,451	9
high-tech**	23	5	2,170	15	6,011	15
Services	151	9	1,737	8	6,135	12
Total	1,746	-	21,869	-	52,281	-

Source: CEPAL (1996)

* % of industrial good exports

** or *diffusers of technical progress* include: capital goods, IT fine-chemicals and biotechnology

As far as industry is concerned, there was a decrease in traditional goods exports: From 76% of total industrial exports, in 1965, to 38%, in 1995, with a correspondent increase of other type of industrial products. However, from 1980 to 1995, only **industrial commodity** exports increased significantly: from 25% to 38%.

Table 5: Brazil: Imports 1965/1995
(in 1995 US\$ millions and percentages)

Imports	1965		1980		1995	
	Value	%	Value	%	Value	%
Primary goods	401	28	12,643	42	7,724	11
energy related	189	13	10,236	34	4,134	6
Industrial goods	695	48	12,305	41	45,996	69
traditional*	67	10	1,024	8	8,490	18
commodities*	352	51	5,802	47	13,886	30
durable*	37	5	495	4	6,904	15
high-tech*	239	34	4,984	41	16,716	37
Services	339	24	4,871	17	13,630	20
Total	1,435	-	29,820	-	67,366	-

Source: CEPAL (1996)

* % of industrial good imports

** or *diffusers of technical progress*, include: capital goods, IT fine-chemicals and biotechnology

Table 5 presents data for imports, organized in the same way as in Table 4. The general trends are, first, the massive import substitution of energy related goods (oil), mainly from 1980 onwards. Second, there is no important change in the imports of services, which account for around 20% of total imports. However, some important changes in industry imports can be noticed, mainly for the 1980-

95 period. First, there was an increase in industrial participation in total imports, from 41%, in 1980, to 69%, in 1995. There were increases in the imports of traditional (8% to 18%) and durable goods (4% to 15%) in the same period. The imports of technical diffuser goods remained in about the same level (around 40%), for the period 1980-1995. The problem here is that the exports of this group of products has been kept at a much lower level than the imports (15%), indicating the fragility of the Brazilian capital goods industry, probably the one that has developed further in Latin America during the *Import Substitution* era.

5. Conclusions: the emerging patterns of Brazilian industry

On the basis of data presented, the emerging patterns of Brazilian industry, at this phase of its adjustment process, can finally be outlined. First of all, it seems that industrial commodities is the group of products in which Brazilian industry has shown some competitiveness, as assessed by the increase in exports and decrease in imports. Therefore, Brazilian international competitiveness is stronger in low unit value, natural resource or energy intensive activities, like pulp, concentrated orange juice, processed soy or semi-processed minerals. Competitiveness relies on resource and infrastructure availability and on technically experienced and efficient firms. Also, these firms enjoy the advantages of double market affiliation, enabling them to shift sales to the internal or external markets, depending on demand prospects. These findings are supported by the evaluation of the market opening policy in Brazil, undertaken by Moreira and Correia (1996), which envisaged in this commodities specialization an efficient resource allocation pattern.

However, such export specialization, while positive for exploring consolidated national advantages, is very sensitive to the evolution of the world economy. For example, the 1993-94 world economic growth implied substantive price

increases for most industrial commodities, offsetting, for Brazilian exporters, the negative impacts of the currency overvaluation induced by *Plano Real*. However, the trends for demand (prices and quantities) are uncertain. These products have low-income elasticity of demand and firms are incapable of influencing price evolution, usually defined in international commodity markets. As a result, while in quantum indices Brazilian exports increased 70%, compared to world trade growth of 40%, international trends towards higher value added led to a gradual erosion of the country's share in world trade from 1.5% in mid-1980s to 0.9% in 1990 (Ferraz, Kupfer and Haguenaue 1996). This trend has not shown any reversal so far.

For equipment and durable goods the situation goes in the opposite direction. The combined influence of instability (followed by stabilization) cum trade liberalization induced imports of finished goods. In response, local producers are specializing in fewer goods with higher levels of imported inputs, relatively to the past. As a consequence, while products became cheaper and technologically updated, benefiting consumers, many manufacturing activities were abandoned and pressures on the trade balance increased, to the point of forcing the government to review the import tariff on cars, which had been set to be 20% in 1994, but stands above 70%.

The prospects for the high-tech (including capital goods) industry are not very promising. The local firms are under pressure from imports and facing an internal market that, as shown by the investment rate, is not presenting signs of demand recovery. Indeed, in 1990, imports of capital goods represented 17.7% of the local consumption, whereas, in 1995, this same indicator reached 41.7% (Mesquita e Correa, 1996). Market concentration, denationalization and downsizing are the main trends for this industrial sector, as exemplified by fine-chemicals: multinational and national companies are closing down their upstream operations and concentrating on the final segments of the market (Teixeira, 1996).

Along the 1980s, under the influence of internal and external economic and institutional forces, industrial firms in Brazil changed. The longevity of the economic instability, while blocking the possibilities for internal expansion, implied more disputes for shares of a shrinking internal market. The external market, always praised by the critics of the import substitution regime, proved to be a partial solution, incapable of pulling the economic growth of a country of continental dimensions, in a sustainable way. In turn, institutional deregulation, trade liberalization and, above all, the pre-eminence of controlling macroeconomic fundamentals, as the objective function of government, implied increasing pressures from internal and external competition.

For the years to come, the adjustment strategies adopted in Brazilian industry will facilitate and, at the same time, impose constraints to the consolidation of economic stabilization and the return of sustainable growth. Higher efficiency certainly contributes to anchor stabilization on sound basis, helping to restore confidence and, hopefully, investment levels. But, if economic concentration prevails and generation of intersectoral and/or spatial externality opportunities is limited, growth prospects may be hindered. Indeed, fiscal and monetary policies are not a substitute for more effective strategies of productivity enhancement. These strategies must center on investing in labor and innovations, stressing the necessity of expanding capacity by investing in new production facilities and hiring workers. Also, if firm and industrial specialization evolves towards low unit value products (relatively to the imported basket), well known but long forgotten problems associated with terms of trade may rise.

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